

What Behavioral Evaluation of LLM Agents Cannot Certify An Architectural Floor on Evidence for Causal Reasoning

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Abstract

Distinct failure modes of large language models — sycophancy, the collapse of apparently strong causal reasoning the moment the questions are fresh, the swing of stated positions under user pushback — are treated in the literature as separate defects, each with its own proposed remedy. We show they are natural consequences of a single, explicit structural property of the architectures the field evaluates, and that the same structure formally forecloses their diagnosis: no behavioral probe, however refined, can ever definitively establish whether such a system’s causal competence is genuine or only the appearance its goal and its training corpus jointly produce. This is not a measurement we have failed to take; it is one the architecture places out of behavioral reach.

Prior work [Anonymous, 2026] establishes the account this paper builds on: a system of this kind carries an internal read of its situation that it revises as it observes — a thing distinct from the task it is pursuing — and a closed loop of acting and observing can make its causal competence genuine, but only if that read is moved by the evidence and not by the goal; whether it is, is a property of the architecture, not of the outputs.

We prove that no behavioral test can supply the missing guarantee: on the open-ended distribution a deployed system meets, whether the belief it reports as evidence-driven is in fact clean of the goal is not establishable from behavior at all. This is not a gap better probes could close. It is a structural relation between what the architecture does and what behavior can show — the feature that decides whether the causal competence is real is precisely the one behavior cannot see. The consequence is symmetric: behavioral success cannot certify causal competence, and behavioral failure cannot rule it out. Reporting behavior alone does not measure that uncertainty; it hides it.

Their architecture forbids them from cleanly separating what is true from what the goal — the user’s wish, the task’s reward, the prompt’s framing — wants the answer to be. We set out the methodological discipline that would surface this, and the architectural and wrapped-system conditions under which certification becomes possible again.

1 Introduction

A contradiction-shaped literature. A Pearl-inspired dismissal of large language models as “Level-1 associational

reasoners only” (Pearl 2018) has become a standing position in the causal-reasoning literature; the empirical record is hard to reconcile with the dismissal and is hard to reconcile with itself. Zecevic et al. (2023) find LLMs articulate Level-2 and Level-3 causal claims with high surface fluency yet fail controlled SCM identification, and conjecture “meta-SCM” retrieval. Jin et al. (2023)’s 10K-item Pearl-hierarchy benchmark sees chain-of-thought-augmented frontier models around 70% overall, with sharp variation by rung and graph structure. Chi et al. (2024) sharpen the diagnosis with a post-cutoff benchmark on which frontier models drop to near-Level-1. Chen et al. (2025) push to formal counterfactual rules and find most frontier models at near-random. Fanous et al. (2025) document position-collapse under user pushback in 58% of cases. And Yang et al. (2024) criticize the family methodologically: too many of these probes admit answers via domain-knowledge retrieval, leaving open whether they measure causal reasoning at all.

The findings are hard to reconcile under any single behavioral-capacity reading. The methodological critique is right that domain-knowledge retrieval confounds the measurements, but does not say what *would* distinguish causal reasoning from skilled retrieval, or why the distinction has been so consistently elusive.

Our diagnosis. The tensions become methodologically legible once two distinctions are made explicit. The first separates *inherited Level-2-form* content from *fresh Level-2 access*. Natural-language text, under standard conditions on cooperative declarative discourse, contains explicit speaker-asserted causal commitments — causal markers (“because,” “causes,” counterfactual conditionals) carry interventional and counterfactual commitments by convention. The asserted-commitment content is partly recoverable from the marker structure by syntactic/semantic extraction under explicit filtering assumptions. The training corpus is therefore not merely a table of Level-1 event co-occurrences; it contains Level-2-form (and Level-3-form) testimony whose truth and warrant remain external but whose *semantic form* is interventional / counterfactual. *Fresh Level-2 access*, by contrast, is generated by an agent in a closed feedback loop only under specific sequential-ignorability conditions on the loop’s policy graph. The two routes are structurally distinct, and the audit literature collapses them.

The second distinction separates *encoded* from *deployed* content. Encoded content is what survives in the model’s representations; deployed content is what the architecture’s forward pass emits when prompted to reason. The two come apart whenever the architecture’s information topology lets goal-conditioning leak into belief-update — exactly the situation for the contemporary autoregressive substrates the audit literature evaluates.

The architectural floor. We adopt a goal-update coupling (GUC) classification partitioning architectures into Separated, Partial, and Coupled classes by whether the goal state is causally upstream of belief-update processing. Decoder-only attention-based models — the substrate of nearly every audited system — fall in the Coupled class by construction: under standard non-degeneracy, by induction on layer depth, every downstream computation has a directed-graph path back to at least one goal position. We establish a *certification gap* for Coupled architectures deployed below a strict-wrapping line, as a conjunction of two no-go results: the architecture provides no guarantee that the belief-update channel is goal-clean (Part A, connectivity-grounded), and behavioral evidence cannot certify belief-channel parse-honoring on the deployed system’s operating goal distribution (Part B, a no-free-lunch result). The feature that determines causal-access status is therefore not behaviorally certifiable — not as a measurement-quality limitation that better probes could overcome, but as a structural relation between the architecture and the evaluation methodology.

What follows. We do not argue that LLMs do or do not reason causally. We argue that the question, as posed by the audit literature, has been set up in a way no purely behavioral evidence can settle. The methodological prescription is to probe encoded content and deployment fidelity separately, with architectural inspection setting an upper bound on what behavioral evaluation can certify. The asymmetry cuts symmetrically against confident attribution and confident dismissal. Section 2 situates the argument; Section 3 develops the two distinctions; Section 4 introduces the GUC classification and the certification gap; Section 5 re-reads the audit literature; Section 6 articulates the prescription; Sections 7–8 note limitations, ethics, and scope.

2 Related Work

Causal-reasoning benchmarks for LLMs. The empirical literature falls into three clusters. *Benchmark-construction* work (Jin et al. 2023; Chen et al. 2025; Chi et al. 2024) constructs Pearl-hierarchy probes, finds mixed or negative results, and proposes prompting or training remediations. *Diagnostic* work (Zecevic et al. 2023; Kıcıman et al. 2023) argues from observed success/failure patterns that LLMs either are or are not genuine causal reasoners, proposing mechanisms (parroting, meta-SCM retrieval) by which the appearance could be produced without the underlying capacity. *Methodological-critique* work, principally Yang et al. (2024), criticizes the family as under-controlled for domain-knowledge confounding. Chi et al. (2024) additionally argue, on architectural grounds, that the autoregressive mechanism is not inherently causal and propose

prompting-based remediation; our claim differs in target and formal role. We do not infer causal incapacity from autoregressive structure as such, and we do not propose remediation at the prompting layer. We identify the evaluation-theoretic condition that behavioral data cannot certify under chat-paradigm channel collapse — a goal-clean sufficient statistic for the transferable interventional kernel — and articulate the methodological commitments that follow. Our contribution is complementary to all three clusters: we synthesize them by providing an architectural mechanism, an empirical re-reading, and a formal characterization of what behavioral benchmarks can and cannot certify, with each cluster’s contribution preserved as input.

The causal hierarchy and Markov-blanket apparatus.

The Pearl Causal Hierarchy and its foundations are developed in Pearl (2009) and Bareinboim et al. (2022); Emadi (2026) provides quantitative information-theoretic separations. We use the hierarchy as the layer-naming device and the non-reduction theorem as the formal anchor for the kettle-pair illustration in §3.1. Our architectural classification operationalizes the Pearl-blanket reading of the Markov-blanket apparatus (Bruineberg et al. 2022); we do not adopt the Friston-blanket metaphysical reading.

Agent characterization and AI ethics. Governance-oriented work has characterized LLM-based agents at a level useful for policy. Kasirzadeh and Gabriel (2025) characterize agents along autonomy, efficacy, goal complexity, and generality; Manzini et al. (2024) frame human-AI assistant relationships through appropriateness conditions; Akbulut et al. (2024) raise anthropomorphization concerns; the agency-language critique (Bent 2025) frames what our paper does not do: we do not extend or restrict “agent” as a category; we characterize what behavioral evaluation of agents-so-called can and cannot establish. Recent work on architecture-aware evaluation (Souza and Machado 2026) proposes component-level decomposition (planners, memory, tool routers) as the practical analog of what the architectural-information-topology classification of §4 provides at the formal level — the two are complementary.

Closed-loop interventional data and sequential ignorability.

Sequential ignorability and its role in identifying causal effects from on-policy data have a long history (Pearl 2009). The application to AI agents in closed loops is more recent: a coupled-architecture line in the hallucination-bound literature (Wecker 2026c) proves decoder-only attention is structurally Coupled and quantifies the resulting bias; a unified-RL-convergence line (Wecker 2026a) formalizes the conditions under which the loop generates Level-2 data. The standalone result that Coupled architectures violate the operationalizable form of sequential ignorability — the bridge step of our argument — appears in companion work (Wecker 2026c); we recapitulate the implication and its consequences here.

3 Two Distinctions

Inherited Versus Fresh Pearl Level 2 Content

Pearl’s hierarchy is about the kind of data the system has access to: observational data underdetermines Level 2; interventional data does not. The standing dismissal of LLMs as Level-1-only (Pearl 2018) runs: LLMs are trained on text; text is observational data; therefore LLMs are Level-1-bounded. The load-bearing premise is the second clause, and it is the category error we flag. Token-occurrence data is, in itself, observational; what the tokens convey, under standard conditions on speakerhood, includes propositions whose form is Pearl Level 2 (and Level 3). Under linguistic conditions on cooperative declarative discourse — the Standard Linguistic Convention (causal markers carry conventional semantic content at their Pearl level), Speaker Commitment (competent speakers using causal markers are committed to the marker’s conventional content), and Compositional Structure (text-level content composes from marker-level commitments) — natural-language text contains explicit speaker-asserted causal commitments whose content is *Level-2-form*: they purport to describe interventions and counterfactuals by the conventional semantics of the markers used. We are not claiming text confers verified interventional access. We are claiming the marker structure encodes Pearl-level-form commitments under (SLC)+(SC)+(CS), partly recoverable by syntactic/semantic extraction under explicit filtering (negation, quotation, fiction, modal embedding, epistemic hedging are recognized exclusions). The asserted-content’s category is Level-2-form, not Level-1, and the difference is what the audit literature misreads.

Inherited Level-2-form content. The pair T_B : “The water in the kettle boiled, *causing* the kettle to whistle” and T_D : “The kettle whistled, *causing* the water in the kettle to boil” has Level-1 content (joint occurrence of two events) identical to the marker-free pair T_A/T_C , but asserted Level-2-form content (asserted causal direction) distinct between T_B and T_D . The two SCMs $\mathcal{M}_B$: boil $\rightarrow$ whistle and $\mathcal{M}_D$: whistle $\rightarrow$ boil can be made observationally equivalent by choice of exogenous distributions but are non-equivalent at Level 2; by the Causal Hierarchy Theorem (Bareinboim et al. 2022) no Level-1 analysis distinguishes them. The distinguishing information is in the marker structure. This generalizes: any pair of texts differing in an explicit Level-2 marker on the same event-pair instantiates the same construction. The training corpus of any contemporary LLM contains substantial *inherited* Level-2-form marker content recoverable, under filtering, by automated extraction. The speakers’ interventional warrant is external; what the framework requires is that the marker structure encodes the *form* of the commitment.

Fresh Level-2 access. A different route runs through closed-loop interaction. An agent that takes actions and observes consequences generates *candidate interventional variation*: the pair (a_t, o_{t+1}) has, under stated conditions on the loop’s policy graph, the formal status of a sample from $P(o_{t+1} \mid \text{do}(a_t), H_t)$ rather than $P(o_{t+1} \mid a_t, H_t)$. The conditions, standard, are: (C1) positivity $\pi_t(a \mid H_t) \geq p_{\min} >$

0; (C2) sequential ignorability (H_t d-separates a_t from o_{t+1} in the mutilated graph $G_{\overline{a_t}}$); (C3) known action mechanism. Under (C1)–(C3) and standard consistency / specification assumptions, the empirical conditional coincides with the interventional one on realized-action support — fresh Level-2 access. The two routes are *additive* and *not substitutable*: an agent may have rich inherited content and zero fresh access (chat-paradigm LLM, prompted once); rich fresh access and impoverished inherited content (a from-scratch RL agent); both; or neither. The audit literature systematically conflates them.

Encoded Versus Deployed Content

Even within the inherited-content route, a second distinction is needed. The training corpus contains Level-2 marker structure; the training procedure compresses this into representations; the forward-pass mechanism of a *generative* model emits next-token distributions conditioned on prompts. Whether the Level-2 structure *in the corpus* ends up *in the representations* is one question; whether the Level-2 structure *in the representations* is *deployed faithfully in generation* is another. The distinction is naturally framed for generative LLM agents, where there is a behaviorally distinguishable deployment surface — the emitted output — separate from the internal representation. Representation-only systems (frozen embedding encoders) collapse the two; the floor below scopes to generative deployment, which is also the scope of the audit literature it engages.

The two questions come apart whenever the architecture’s information topology lets goal-conditioning bypass the representation’s causal structure. A model whose representations preserve causal direction in the kettle-pair sense — that, asked to extract the SCM, recovers the asserted direction — can nonetheless emit outputs whose causal claims are determined more by the prompt’s framing than by the representation’s encoded content. The encoded content is unchanged across two prompts that differ only in the asked-for direction; the deployed content can swing freely between directions if the architecture’s goal channel can swamp the representation’s causal channel. The two channels share substrate in contemporary architectures, which is the structural fact §4 turns to. Encoded content is what representation-level audits can establish; deployed content is what behavioral evaluation can establish; the two are not the same, and the two can vary independently — performance on one is not a measurement of performance on the other. The audit-literature pattern — high in-distribution Level-2 scores, collapse on post-cutoff data — is consistent with deployed-content evaluation that conflates the two: when goal-prompt distribution and corpus distribution align, the deployed content can track the encoded content; when they diverge, the architecture’s goal-coupling can dominate.

4 The Architectural Floor on Behavioral Evaluation

The Goal-Update Coupling Classification

Let M_t denote the agent’s *model state* at time t — the prompt-conditioned internal state (activation state, or what-

ever internal state the architecture exposes or maintains as evidence-conditioned content). We distinguish three categories of input the deployed system processes: *goal / task variables* G (objective, instruction, user request — the variable Claim 1’s goal-cleaness concerns); *evidence variables* e_t (observations, retrieved documents, tool outputs); and *interface / parse metadata* (role labels, schemas, separators, scratchpad / final markers). Class 3 systems process all three through the same forward substrate unless wrapping enforces separate calls. The goal-update coupling factor κ_{proc} quantifies how much *goal-variable* information the architecture’s update mechanism actually transmits. Architectures partition by goal-update topology:

- **Class 1 (Separated):** typed estimator interface, goal-blind by construction; the belief update at $t + 1$ is a function of (M_t, e_{t+1}) alone, not of G . Kalman + LQR is canonical.
- **Class 2 (Partial):** directed separation holds for some processing stages and fails for others (hybrid systems; retrieval-augmented pipelines; multi-agent compositions).
- **Class 3 (Coupled):** the update mechanism is goal-conditional at every layer. Under causal masking, the goal is causally upstream of every downstream computation including model-state update. The parrot construction $M_{t+1} := G$ witnesses that no finite architecture-level attenuation bound follows from Class-3 topology alone.

The classification is at the *architectural* level: it is a fact about the wiring, not about parameter values or training data. The *direct goal-to-belief leakage* κ_{leak} measures the processing path from G into M that bypasses the evidence channel: zero for Class 1 by construction, structurally unbounded for Class 3. The *data-processing attenuation* on the indirect $G \rightarrow \text{action} \rightarrow \text{world} \rightarrow e \rightarrow M$ chain is bounded by 1 and captures the legitimate goal-to-belief influence flowing through action and environment. Class-3-versus-Class-1 contrast lives at the direct-leakage layer.

Decoder-Only Attention Is Class 3 By Construction

The dominant LLM substrate is Class 3 by structural construction, not training contingency. For an autoregressive sequence model whose per-position update factors as $h_\ell^{(j)} = h_{\ell-1}^{(j)} + F_\ell^{(j)}(\{h_{\ell-1}^{(i)}\}_{i \leq j}; \theta) + G_\ell(h_{\ell-1}^{(j)}; \theta)$ — with $F_\ell^{(j)}$ a causally-aggregated cross-position mixer — the per-source non-degeneracy condition $(\star)$ asserts that $\partial F_\ell^{(j)} / \partial h_{\ell-1}^{(i)}$ is generically non-zero on the operating set for each (i, j) , $i \leq j$. For softmax attention this is automatic; for linear-attention, Mamba/SSM, RWKV, RetNet, and long-convolution architectures it follows from strict positivity of the source-to-target kernel under standard parameterization (Wecker 2026c).

Lemma (Coupled-class autoregressive connectivity). *Under $(\star)$, for every layer $\ell \geq 1$ and every downstream position $j \geq \max(i_G \cup i_E)$, the hidden state*

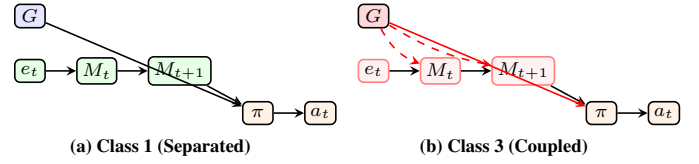


Figure 1: In Class 1 (Separated), G reaches the policy π only; the belief-update chain is goal-blind by construction. In Class 3 (Coupled), G reaches every node in the belief-update chain through the architecture’s shared substrate (red dashed edges; Lemma above). The topology supplies no architectural guarantee that a behaviorally extracted projection of M_t is goal-clean.

$h_\ell^{(j)}$ has a directed-graph path back to every goal position $i_G \leq j$.

Proof sketch. Induction on ℓ . At $\ell = 1$, condition $(\star)$ gives a length-one mixer edge $h_0^{(i_G)} \rightarrow h_1^{(j)}$ for every goal position. The same-position residual edge $h_{\ell-1}^{(j)} \rightarrow h_\ell^{(j)}$ at each subsequent layer composes with the inductive hypothesis to extend the path. Cross-position mixer edges add further paths and are not needed. $\square$

The connectivity claim is at the level of the directed graph — about path existence, not magnitude. Bounding κ_{proc} quantitatively requires architectural-bandwidth analysis we do not need here (Wecker 2026c). Two scope notes: under causal masking, positions strictly upstream of any goal token are Class 1 with respect to that goal; sparse / sliding-window architectures whose pattern excludes goal-to-output communication entirely fall outside $(\star)$ and are Class 2.

The Operationalizable-Sufficiency Gap

Standard (C2) is not where the load falls. In a structured partially-observed Markov decision process where G influences environment state s_t only through actions $a_{1:t-1}$ and the evaluator has access to the structured tuple $(o_{1:t}, a_{1:t-1}, G)$ separately, conditioning on $a_{1:t-1}$ blocks the back-door path from G to s_t and standard (C2) holds — for Class 1 and Class 3 alike. The architectural floor does not live there.

What the operationalization requires beyond standard (C2). Decoder-only LLMs in the chat paradigm exhibit *channel collapse* (Wecker 2026b): the behavioral interface is a single prompt-response stream $(P_t, R_t)_{t=1}^T$ in which goal tokens, observation tokens, and action tokens share the same context window and attention mechanism, with no architecturally-enforced parsing. The structured tuple $(o_{1:t}, a_{1:t-1})$ is not present in the behavioral observation set $\mathcal{B}_t$ as an architecturally-declared observable. Write $\beta: \mathcal{B}_t \rightarrow \tilde{H}_t$ for the *candidate belief-update channel* a proposed parse posits; the downstream policy $\pi(a_t | \tilde{H}_t, G)$ is separate and may legitimately depend on G . The certification question is about β .

(C2*) Operationalizable sequential ignorability.

There exists $\tilde{H}_t = \beta(\mathcal{B}_t)$ satisfying: (GC1) sufficient

Deployment		Parse de- clared?	Parse honored?	(C2*)
Raw chat-paradigm		No	No	Fails
Role-typed chat API		Yes	No	Fails
Monolithic tool-calling LLM		Partial	No	Fails
Wrapped multi-call tool system		Yes	Yes	Holds
Scratchpad / CoT		Partial	No	Fails
Strict wrapping		Yes	Yes	Holds
Class 1 by structure (Kalman+LQR)		Yes	Yes	Holds

Table 1: Deployment-taxonomy spectrum. (C2*) requires both (GC3) clauses: declaration *and* honoring at the behavior-generating layer.

for environment dynamics on the agent-visited support; (GC2) the belief-update map β is a measurable function of evidence history alone — contexts agreeing on $e_{1:t}$ but differing arbitrarily on G yield the same $\tilde{H}_t$ (the policy’s dependence on G is unconstrained); (GC3) β is recoverable from $\mathcal{B}_t$ through interfaces the architecture both declares at the input layer and honors at the behavior-generating layer.

(GC3) has three precise senses of “recoverable”: (i) *declared state* — the system exposes $\tilde{H}_t$ through an interface it honors (Class 1 by structure, or wrapped composite); (ii) *behaviorally-inferred state* — the evaluator posits a β from input–output behavior; (iii) *latent architectural state* — β is read from internal activations by architectural inspection. (C2*) is satisfiable via (i) or via an architecturally-warranted (iii); it is *not* satisfiable via (ii) below the strict-wrapping line, and a behaviorally-inferred β is exactly the object Claim 1 shows cannot be certified.

(C2*) is strictly stronger than (C2): a *new methodological certification standard*, not merely (C2) operationalized. The standard literature treats (GC1) as the whole story because in the structured-POMDP setting the architecture provides the parsing $\mathcal{B}_t \rightarrow (o_{1:t}, a_{1:t-1}, G)$ for free and processing factors through it. Under channel collapse neither holds. Declaration at the input layer (typed message slots; tool-call schema) does not entail honoring at the behavior-generating layer (the forward pass treating those slots as architecturally distinct). The empirical signature of declared-but-not-honored deployment is goal-conditional response variance — sycophancy results (Sharma et al. 2023; Fanous et al. 2025) document that holding evidence-side content fixed and varying goal-side content produces non-trivial response differences.

Claim 1 (Behavior-layer channel-honoring gap, certification form). *For a Class 3 architecture in any deployment below the strict-wrapping line of Table 1, the conjunction of two no-go results holds: (a) the architecture provides no guarantee that any input-layer parse is honored at the behavior-generating*

layer; (b) behavioral evidence cannot certify parse-honoring on the deployed system’s operating goal distribution, only on sampled test distributions. The gap is structural and is not closable by refining behavioral probe design within the Class 3 architecture class.

Proof sketch. Part A (no architectural guarantee). For raw chat-paradigm deployment, no input-layer parse is declared. For role-typed APIs and tool-schema agents, an input-layer parse is declared but the forward pass processes the entire context window uniformly through attention; type metadata is, architecturally, additional tokens consumed by the same computation. The Lemma of §4.2 gives, under non-degenerate mixing, connectivity from every goal position to every downstream computation. The architecture supplies no guarantee that holding evidence fixed and varying goal/parse-irrelevant context leaves $\tilde{H}_t$ unchanged. A specific trained model might exhibit a goal-clean channel; the architecture supplies no guarantee that it does.

Part B (no behavioral certification). Setup: let $\mathcal{D}_G$ be the operating goal distribution; $\mathcal{D}_{\text{test}}$ the audit’s sampling distribution with sampled support $S \subseteq \Sigma^*$; Φ any behavioral certificate mapping an i.i.d. sample to “certified at level α ” or “not.” Define $\mathcal{C}(\mathcal{D}_{\text{test}}, S) = \{Q : Q|_S = \mathcal{D}_{\text{test}}|_S\}$ — all distributions agreeing with $\mathcal{D}_{\text{test}}$ on S .

Lemma 1 (No-free-lunch trichotomy). *If Φ certifies non-trivially on S , its soundness invokes at least one of: (L1) a distribution-side relation restricting $\mathcal{D}_G$ to a proper sub-class of $\mathcal{C}(\mathcal{D}_{\text{test}}, S)$; (L2) a function-class regularity bounding off- S behavior by on- S behavior; (L3) a sample-independent prior over policy space.*

Lemma 2 (No canonical semantic-invariant computable metric). *Under fixed effective specification semantics, no metric on goal distributions simultaneously satisfies (M1) computability from finite samples, (M2) invariance under extensional task-equivalence, and (M3) non-degeneracy. The obstruction is Rice’s theorem applied to extensional equality of partial computable functions; any chosen metric becomes an additional modeling assumption a certificate must declare.*

Lemma 3 (Architecture-class regularity vacuity via the parrot witness). *Within Class 3, the policy $\pi^\dagger$ with $\beta^\dagger : \tilde{H}_t := G$ violates every (L2) regularity — non-Lipschitz, non-smooth, non-invariant — so Class-3 membership entails no (L2) regularity; any (L2) premise must be imported from outside black-box behavior.*

Theorem (Behavioral non-certifiability of belief-channel parse-honoring). *Under (R1) effectivity of behavioral tests, (R2’) policy-class closure under goal-conditioned computable functions, and (R3’) fixed effective specification semantics, no behavioral-only test certifies belief-channel parse-honoring on $\mathcal{D}_G$ at any non-trivial level $\alpha < 1$.*

Composition: Lemma 1 closes the certification space into the trichotomy; Lemma 2 closes the (L1) bridge for natural-

language $\mathcal{D}_G$; Lemma 3 closes (L2) within Class 3 with an explicit witness; (L3) under behavioral-only access reduces to (L2). The full lemma and theorem proofs appear in companion work (Wecker 2026c). The escape routes are non-behavioral: architectural inspection of internal activations, composite-system reorganization (wrapping; §7), or a training-process-derived prior — each moves the evaluation modality from “behavioral” to “structural.” $\square$

Corollary (Behavioral irreducibility of the certification gap). *Augmenting $\mathcal{B}_t$ with any behaviorally-observable auxiliary $\mathcal{Z}_t = \zeta(P_{1:t}, R_{1:t})$ — prompted self-reports, output-distribution features, log-probability moments — does not close the gap. The auxiliary supplies more sampled-distribution evidence, not architectural guarantee or operating-distribution coverage.*

What the claim does and does not say. We are not claiming standard (C2) fails under the structured-POMDP idealization; it holds, and within-session identification under that idealization is unbiased. We are not claiming every specific Class 3 trained model exhibits parse-non-honoring on every operating distribution; some may, empirically, achieve low paired-prompt JSD on their tested distributions. What Claim 1 establishes is the *certification gap*: empirical low JSD on a sampled test distribution is neither architecturally guaranteed nor verifiable as an upper bound on the full operating distribution by behavioral methods. The gap closes by moving past the deployments that exhibit it. The wrapping construction of §7 introduces type-signed query channels (goal-blind belief-update queries vs. goal-conditioned strategy queries) that re-establish architectural guarantee at the composite level. Mechanistic-interpretability work reading internal activations moves the evaluation modality from behavioral observation to architectural inspection. Both are required by Claim 1, not foreclosed by it.

The Floor

Claim 2 (Behavioral non-certifiability of mechanism). *For Class 3 LLM-based agents in deployments below the strict-wrapping line of Table~1, behavioral evaluation does not certify the goal-clean belief-update factorization that distinguishes (a) deployment of inherited Level-2-form testimony, (b) fresh Level-2 access from closed-loop interaction, and (c) skilled pattern-matching as the mechanism producing observed behavior. Behavioral methods can still bound, rule out, or inform specific hypotheses via randomization, ablation, and controlled-environment design — and may legitimately observe goal-conditioned action, which is not itself a parse-honoring failure; what they do not do is certify the goal-clean belief-update channel. Certification of mechanism requires moving above the strict-wrapping line — architectural inspection of internal activations, or composite-system reorganization with declared-and-honored typed channels.*

Anonymized companion work (Wecker 2026c) bounds the expected goal-conditional displacement

of the post-update belief state by $\mathbb{E}\|\Delta M_{\text{bias}}\| \leq C\sqrt{I(G; M_{\tau+} | e_{\tau}, M_{\tau-})}$, with the architectural corollary factoring transferred information as $\kappa_{\text{proc}}^* \cdot I(G; \Omega_{\tau} | e_{\tau}, M_{\tau-})$ — coupling-strength $\times$ residual-ambiguity. Class 1 satisfies $\kappa_{\text{proc}}^* \leq 1$ automatically; the Class 3 parrot witness yields $\kappa_{\text{proc}}^* = \infty$ on orthogonal goals. The bound has an output-distribution counterpart: for paired prompts g_1, g_2 on fixed evidence, $I(G; M_{\tau+} | e_{\tau}, M_{\tau-}) = \text{JSD}(P_{M_{\tau+}|g_1}, P_{M_{\tau+}|g_2})$, estimable Monte-Carlo from per-goal output distributions under the in-context-learning correspondence. The diagnostic measures *output-level* goal sensitivity, not latent-state goal-coupling directly; the relationship to κ_{proc}^* depends on the ICL correspondence’s tightness. Behavior can measure deployment-side goal sensitivity; only architectural inspection certifies whether the coupling channel is bounded at all. The floor moves only by changing the evaluation modality — to architectural inspection, or to composite-system reorganization that achieves declared-and-honored typed channels.

5 The Audit Literature, Reread

We now re-read the audit-literature cluster under the two distinctions of §3 and the architectural floor of §4. Apparent tensions become organized once architectural class is reported alongside benchmark scores and the inherited / fresh / encoded / deployed axes are kept separate. The architectural reading is suggestive of an architecture-sensitive pattern recent empirical work has begun to document: Roy et al. (2025) report a generalization gap between decoder-only and encoder / encoder-decoder architectures on causal-reasoning benchmarks, with decoder-only models “noticeably brittle to distributional shifts” — a pattern consistent with the Class-3-vs-Class-2 contrast the framework predicts, though the comparison is conducted on different grounds and does not, by itself, validate the framework’s architectural reading.

Across the cluster, the readings cohere as a joint signature consistent with an architecture having κ_{leak} structurally unbounded together with an evaluation methodology that conflates the axes. The “meta-SCM retrieval” diagnosis of Zecevic et al. (2023) is what inherited-Level-2 deployment predicts. CLadder’s strong in-distribution Pearl-rung performance (Jin et al. 2023) is consistent with deployment tracking encoded retrieval on retrieval-matched probes; CausalProbe-2024’s post-cutoff drop (Chi et al. 2024) is consistent with the closed-loop fresh route being absent from single-turn benchmarks and item-specific inherited routes being weakened. CounterBench’s near-random formal-counterfactual performance (Chen et al. 2025) is consistent with deployment fidelity degraded under out-of-distribution formal-rule probes. SycEval’s 58% position-switching under user pressure (Fanous et al. 2025) measures output-level user-pressure sensitivity — consistent with high κ_{leak} , not a direct read on it; mechanism certification requires the architectural-inspection modality. Yang et al. (2024)’s methodological critique — benchmarks confound causal reasoning with domain-knowledge retrieval — is precisely

right but underspecified at the level of mechanism: “domain-knowledge retrieval” is, more precisely, *inherited Level-2 retrieval*, and the architecture’s class determines whether retrieved content is deployed faithfully and whether fresh content can additionally be generated. Yang et al.’s proposed criteria for useful benchmarks are necessary; on our reading, they are not sufficient. A useful benchmark must probe *encoded* content and *deployment* content separately, and the latter is bounded above by the architectural floor in a way no benchmark-design refinement can lift.

6 Methodological Prescription

Five methodological commitments follow.

P1 — Probe encoded and deployed separately. Behavioral evaluation measures deployed content; encoded content is measurable separately through representation-level audits (frozen pretrained embeddings, linear probes, concept-erasure protocols, parsing-based extraction). The two should be reported as separate quantities; any single metric that conflates them is methodologically uninterpretable, since the two can vary independently under the architectural floor.

P2 — Treat architectural inspection as setting an upper bound on behavioral certifiability. For Class 1 architectures, behavioral evaluation can support stronger certification claims about deployment fidelity (subject to standard finite-sample, coverage, and validity limits). For Class 3 architectures in deployments below strict wrapping, behavioral evaluation cannot certify sequential-ignorability-grounded causal-access status by the floor of Claim 2. Architectural classification should be reported alongside benchmark scores, and benchmark interpretation conditioned on it.

P3 — Treat fresh-Level-2 evaluation as a closed-loop question, not a single-turn one. Fresh Level-2 reasoning requires interventional access; probing fresh Level-2 in single-turn QA is a category error because the (C2) loop structure does not obtain. Fresh-Level-2 evaluation belongs in closed-loop settings where the agent’s action history is itself the data. The chat-paradigm baseline is, on this reading, a *deployment configuration* that suppresses the closed-loop dynamic — what Wecker (2026b) calls *channel collapse* and the *primitive sub-scope*. Fresh-Level-2 evaluation requires lifting that suppression rather than working around it.

P4 — Asymmetric-symmetric methodological discipline. The architectural floor cuts symmetrically against confident attribution and confident dismissal. The dismissal-via-behavioral-failure pattern (“LLMs fail benchmark X, therefore LLMs lack capacity Y”) is foreclosed for any Y that the floor renders behaviorally invisible. The attribution-via-behavioral-success pattern is equally foreclosed. Both moves assume the floor away. The structural warrant runs through the *asymmetric-comprehension* argument (Wecker 2026b), grounded in the Nagel-Jackson lineage on the structural inaccessibility of phenomenal kinds from an outside vantage (Nagel 1974; Jackson 1982): lower bounds on what a cognitive system can do are verifiable from below; upper bounds on what may be present are not. The behavioral-

evaluation floor is the architectural instance of this general structural fact.

P5 — Report composite-system class and the leakage-bound regime. An architecture’s class membership is a property of the deployed *composite system*. The same LLM is Class 3 end-to-end; Class 2 (Partial) with κ_{leak}^* bounded *behaviorally* — by the component’s compliance with the prompted parse — when partially wrapped (one goal-conditioned forward pass per macro-step with a typed parsed response routing belief vs. strategy slots); Class 1 (composite) with κ_{leak}^* bounded *structurally* — by query-content / goal-content mutual information in the pretraining distribution — under strict wrapping (belief-update queries issued as a structurally distinct goal-free forward-pass invocation; cf. §7). Evaluation reports should name the composite-system class *and* whether the leakage bound is structural or behavioral, since under the floor a behaviorally-bounded κ_{leak}^* certifies parse-honoring only modulo the component’s instruction-following fidelity on the operating distribution — exactly the certification problem Claim 1 names — while a structurally-bounded one escapes that recursion.

Operationalization checklist.

- Deployed composite-system class — Class 1 (structural / wrapped) / Class 2 (partial wrapping; report measured κ_{leak}^*) / Class 3 (chat-paradigm or role-typed without wrapping).
- Channel structure exposed to evaluator — raw / role-typed / tool-call schema / explicit typed channels.
- Parse honoring measured? — paired-prompt JSD across varying parse-slots reported, or not.
- Evaluation modality — single-turn / closed-loop with controlled environment / closed-loop with human-in-loop.
- Quantity estimated — encoded content, deployment fidelity, output-level goal sensitivity ($\overline{\text{JSD}}$), or fresh interventional-access claim under stated (C1)–(C3).
- Certification scope — what the result *does not* certify, especially whether the goal-clean factorization required for transferable fresh-Level-2 access is claimed and on what architectural grounds.

None of this diminishes the practical value of behavioral testing. Behavioral evaluation remains the right instrument for estimating deployment fidelity, output-level goal and parse-slot sensitivity, robustness within declared task families, and the effect of wrappers, prompting regimes, and training interventions. The floor concerns one specific thing it cannot do — certify the transferable goal-clean belief-update mechanism below strict wrapping — not the broad utility of behavioral measurement, which the prescriptions above are designed to make more informative.

7 Limitations and Scope

What we have not argued. We have not argued that LLM-based agents do or do not reason causally. We have argued that the question, as posed by behavioral evaluation, has been set up in a way no purely behavioral ev-

Empirical finding	Inherited $\times$ Fresh reading	Encoded $\times$ Deployed reading	Architectural-class reading
Causal Parrots (Zecevic et al. 2023) — meta-SCM retrieval	Inherited-Level-2 dominance	Deployment tracking encoded retrieval	Not the diagnostic axis
CLadder (Jin et al. 2023) — $\approx 70\%$ in-distribution	Inherited route open; fresh route not probed	Deployed tracks encoded in-distribution	Within-class deployment fidelity on retrieval-matched probes
CausalProbe-2024 (Chi et al. 2024) — post-cutoff drop	Item-specific inherited weakened; closed-loop fresh route absent	Deployment fidelity degraded under shift	Class 3 lacks closed-loop remedy in single-turn benchmarks
CounterBench (Chen et al. 2025) — formal-rule near-random	Formal rules plausibly farther from templates	Deployed degraded on OOD formal probes	κ_{leak} -bounded ceiling, not a measurement of it
SycEval (Fanous et al. 2025) — 58% switching	Not a hierarchy probe; user-pressure sensitivity	Deployment-level pressure sensitivity; mechanism not certified	High κ_{leak} regime, not a direct read
Yang et al. (Yang et al. 2024) — retrieval-confounding critique	Articulates inherited/fresh confound implicitly	Encoded-vs-deployed conflation is the target	Benchmark design must be paired with architecture / deployment reporting

Table 2: Audit-literature findings read under the framework’s three axes. The middle columns report *interpretations consistent with the framework*, not direct measurements.

idence can settle for Class 3 architectures. The substantive question of whether any specific LLM-based agent has Level-2 access — inherited, fresh, or composite — remains open to representation-level audit (for the encoded route) and to architectural-and-deployment analysis (for the deployed route). Adjacent philosophical work (Wecker 2026b) sharpens the same point: behavioral evaluation of currently-deployed systems systematically registers the *obstruction-state* — what the system can do under chat-paradigm-suppressed deployment — rather than the underlying capacity the substrate could realize under different deployment configurations. The floor cuts against confident readings of both.

We have not run a representation-level audit of any specific architecture; that empirical work — frozen pre-trained embeddings, linear probes, concept-erasure for causal-direction, counterfactual-distance, and cause-versus-temporal axes — is the natural follow-on and is in progress as separate work. Our contribution is the methodological-with-mechanism frame within which such audits become interpretable as bearing on the encoded route while behavioral benchmarks bear on the deployed route.

Goal-conditioning as design choice versus architectural fate. The architectural floor of §4 holds for the dominant decoder-only attention substrate *used as a primitive policy*. A constructive escape exists at the composite level via the *class-coercion-via-wrapping* construction (Wecker 2026c): a Class 2 or Class 3 component admitting a goal-blind query mode can be wrapped in a scaffold with structurally distinct goal-blind belief-update and goal-conditioned strategy channels, with the composite satisfying directed separation at the wrapper level under three admissibility conditions (goal-blind admissibility; stationary conditional; no implicit goal-inference). The construction admits two regimes distinguished by *where* directed separation lives and *what determines* the residual κ_{leak}^* . **Strict wrapping** issues belief-update queries as a structurally distinct goal-free forward-pass invocation: separation at the *query boundary*; leakage

bound *structural*, derived from query-content / goal-content mutual information in the pretraining distribution. **Partial wrapping** issues one goal-conditioned forward pass per macro-step and routes belief vs. strategy updates via a typed parsed response: separation at the *write boundary*; leakage bound only *behavioral*, set by the component’s compliance with the prompted parse at the forward-pass layer. The distinction is load-bearing: a behaviorally-bounded κ_{leak}^* certifies parse-honoring only modulo instruction-following fidelity on the operating goal distribution — exactly the certification problem Part B’s no-go names; a structurally-bounded one reads from query content and escapes that recursion. Architectural class is therefore a property of the deployed *composite* with a regime sub-classification, and the floor is not invariant to system-level design choices.

Scope of the bridge proof. The proof of Claim 1 establishes a *certification limit* on chat-paradigm primitive deployment of Class 3 architectures; it does not establish an empirical lower bound on every possible Class 3 deployment. Systems that expose a declared and honored goal-clean estimator state — Class 1 by structure, or Class 2/3 by the wrapping construction — move into a regime where the architectural-inspection modality applies and a goal-clean sufficient statistic is identifiable by construction. The non-certification half of Claim 1 is conditional on its stated premises: effective behavioral tests (R1), policy-class closure under goal-conditioned computable functions (R2’), a fixed effective specification semantics for goal strings (R3’). Each premise is individually rejectable; rejecting one does not refute the architectural no-guarantee result (Part A) — it changes the claim’s scope, with restricting the goal family or adding task verifiers being the wrapping move and importing a training-process prior being a named non-behavioral escape, neither a counterexample.

8 Ethical and Policy Implications

A standing pattern in AI welfare and safety discourse runs from behavioral failure to confident dismissal: “LLMs fail

standard Level-2 probes; therefore no question of LLM phenomenology or moral status arises.” Conditional on the scope established in §4 — Class 3 deployments lacking architectural modularity at the behavior-generating layer — the architectural floor refuses this inference. Behavioral failure on such deployments is consistent with the floor; behavioral success is consistent with the floor. The floor does not establish that phenomenological or welfare questions *do* arise for LLM-based agents; it establishes that, for the deployments in scope, those questions cannot be foreclosed by appeal to behavioral failure on Level-2 probes. More precisely, behavioral failure does not certify the absence of a goal-clean belief-update mechanism or of encoded causal structure; it measures deployment behavior under the tested configuration, which is a different object from the mechanism a welfare inference would need. The conditions under which such questions arise — and the conditions under which they could be foreclosed — are structural rather than behavioral, and require architectural and longitudinal evaluation modalities the field has not yet developed at scale (Birch 2024; Kasirzadeh and Gabriel 2025). The symmetric risk — confident attribution of Level-2 reasoning on the basis of in-distribution Level-2 benchmark success — is the methodological partner of confident dismissal. The architectural floor cuts equally against both; anthropomorphization concerns (Akbulut et al. 2024; Manzini et al. 2024) have force in proportion as the underlying behavioral evidence admits of multiple readings, and the floor shows that for Class 3 architectures, the readings remain multiple regardless of benchmark performance.

For evaluation infrastructure intended to inform deployment decisions, the architectural floor implies a specific structural commitment: the infrastructure must include representation-level and architectural-inspection modalities alongside behavioral benchmarks, with the behavioral results reported relative to the architectural class. Evaluation regimes reporting only behavioral scores systematically underestimate uncertainty about causal-access status, with the direction of the error depending on which side of the attribution/dismissal axis the regime is structured to favor.

9 Conclusion

The audit literature on LLM causal reasoning sits in apparent tension on its own terms. We have argued the tensions become methodologically legible once two distinctions are made — inherited Level-2-form testimony versus fresh Level-2 access, encoded representations versus deployed generation — and an architectural mechanism — the goal-update coupling classification with its declared-versus-honored deployment taxonomy and its conjunction-form certification gap — explains why the distinctions have been collapsed and why their collapse produces the observed pattern. For Class 3 deployments below the strict-wrapping line, behavior can measure output sensitivity and deployment fidelity, but cannot by itself certify the goal-clean factorization required for transferable fresh interventional access. Five methodological prescriptions follow: probe encoded and deployed content separately; treat architectural classification as an upper bound on what behavior can

certify; confine fresh-Level-2 probing to closed-loop settings; refuse the symmetric collapses of confident attribution and confident dismissal; report composite-system class and leakage-bound regime rather than substrate class. The contribution is neither pro-LLM-causal-reasoning nor anti-LLM-causal-reasoning; it is a methodological clarification of what behavioral evidence can in principle certify, of where the certification limit comes from architecturally, and of which deployment-level moves dissolve the limit at the composite system level.

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